(https://databricks.com)

Ingest circuits.csv file

```
dbutils.widgets.help()
```

dbutils.widgets provides utilities for working with notebook widgets. You can create different types of widgets and get their bound value. For more info about a method, use **dbutils.widgets.help("methodName")**.

combobox(name: String, defaultValue: String, choices: Seq, label: String): void -> Creates a combobox input widget with a given name, default value and choices
dropdown(name: String, defaultValue: String, choices: Seq, label: String): void -> Creates a dropdown input widget a with given name, default value and choices
get(name: String): String -> Retrieves current value of an input widget
getAll: Map -> Retrieves a mapping of all current values of the input widgets
getArgument(name: String, optional: String): String -> (DEPRECATED) Equivalent to get
multiselect(name: String, defaultValue: String, choices: Seq, label: String): void -> Creates a multiselect input widget with a given name, default value and choices
remove(name: String): void -> Removes an input widget from the notebook
removeAll: void -> Removes all widgets in the notebook
text(name: String, defaultValue: String, label: String): void -> Creates a text input widget with a given name and default value

```
dbutils.widgets.text("p_data_sourec", "")
v_data_sourec = dbutils.widgets.get("p_data_sourec")
```

```
dbutils.widgets.text("p_file_date", "2021-03-21")
v_file_date = dbutils.widgets.get("p_file_date")
```

```
%run "../includes/configuration"
```

```
%run "../includes/common_functions"
```

either above or below command with full path will work

```
# %run /Workspace/Formula1/includes/configuration
```

```
# Databricks notebook source
raw_folder_path = "/mnt/formula1dl1204/raw"
processed_folder_path = "/mnt/formula1dl1204/processed"
presentation_folder_path = "/mnt/formula1dl1204/presentation"

#raw_folder_path = 'abfss://raw@formula1dl.dfs.core.windows.net'
#processed_folder_path = 'abfss://processed@formula1dl.dfs.core.windows.net'
#presentation_folder_path = 'abfss://presentation@formula1dl.dfs.core.windows.net'
```

Step1 - read the csv file using spark dataframe reader

```
display(dbutils.fs.mounts())
```

Table

	mountPoint	source	encryptionType
1	/databricks-datasets	databricks-datasets	
2	/mnt/formula1dl1204/presentati...	abfss://presentation@formula1dl1204.dfs.core.windows.n...	
3	/mnt/formula1dl204/demo	abfss://demo@formula1dl1204.dfs.core.windows.net/	
4	/mnt/formula1dl1204/processed	abfss://processed@formula1dl1204.dfs.core.windows.net/	
5	/Volumes	UnityCatalogVolumes	
6	/mnt/formula1dl1204/raw	abfss://raw@formula1dl1204.dfs.core.windows.net/	
7	/databricks/mlflow-tracking	databricks/mlflow-tracking	
8	/databricks-results	databricks-results	
9	/databricks/mlflow-registry	databricks/mlflow-registry	
10	/Volume	DbfsReserved	
11	/mnt/formula1dl1204/demo	abfss://demo@formula1dl1204.dfs.core.windows.net/	
12	/volumes	DbfsReserved	
13	/	DatabricksRoot	
14	/volume	DbfsReserved	

14 rows

```
%fs
ls /mnt/formula1dl1204/raw
```

Table

	path	name	size	modificationTime
1	dbfs:/mnt/formula1dl1204/raw/2021-03-2...	2021-03-21/	0	1721668310000
2	dbfs:/mnt/formula1dl1204/raw/2021-03-2...	2021-03-28/	0	1721668310000
3	dbfs:/mnt/formula1dl1204/raw/2021-04-1...	2021-04-18/	0	1721668310000

3 rows

```
from pyspark.sql.types import StructType, StructField, IntegerType, StringType, DoubleType
```

```
circuits_schema = StructType(fields = [StructField("circuitId", IntegerType(), False),
                                       StructField("circuitRef", StringType(), True),
                                       StructField("name", StringType(), True),
                                       StructField("location", StringType(), True),
                                       StructField("country", StringType(), True),
                                       StructField("lat", DoubleType(), True),
                                       StructField("lng", DoubleType(), True),
                                       StructField("alt", IntegerType(), True),
                                       StructField("url", StringType(), True)
 ])
```

```
# circuits_df = spark.read \
#     .option("header", True) \
#     .option("inferSchema", True) \
#     .csv(f"{raw_folder_path}/{v_file_date}/circuits.csv")
```

```
circuits_df = spark.read \
    .option("header", True) \
    .schema(circuits_schema) \
    .csv(f"{raw_folder_path}/{v_file_date}/circuits.csv")
```

```
# type(circuits_df)
```

```
# circuits_df.show()
```

```
display(circuits_df)
```

Table

	circuitId	circuitRef	name	location	country	lat	lng
1	1	albert_park	Albert Park Grand Prix Circuit	Melbourne	Australia	-37.8497	144.96
2	2	sepang	Sepang International Circuit	Kuala Lumpur	Malaysia	2.76083	101.73
3	3	bahrain	Bahrain International Circuit	Sakhir	Bahrain	26.0325	50.510
4	4	catalunya	Circuit de Barcelona-Catalunya	Montmeló	Spain	41.57	2.2611
5	5	istanbul	Istanbul Park	Istanbul	Turkey	40.9517	29.40
6	6	monaco	Circuit de Monaco	Monte-Carlo	Monaco	43.7347	7.4205
7	7	villeneuve	Circuit Gilles Villeneuve	Montreal	Canada	45.5	-73.522
8	8	magny_cours	Circuit de Nevers Magny-Cours	Magny Cours	France	46.8642	3.1636
9	9	silverstone	Silverstone Circuit	Silverstone	UK	52.0786	-1.0169
10	10	hockenheimring	Hockenheimring	Hockenheim	Germany	49.3278	8.5658
11	11	hungaroring	Hungaroring	Budapest	Hungary	47.5789	19.248
12	12	valencia	Valencia Street Circuit	Valencia	Spain	39.4589	-0.33166
13	13	spa	Circuit de Spa-Francorchamps	Spa	Belgium	50.4372	5.9713
14	14	monza	Autodromo Nazionale di Monza	Monza	Italy	45.6156	9.2811

77 rows

```
# circuits_df.printSchema()
```

```
# circuits_df.describe().show()
```

Step2 - Select only the required columns

```
circuits_selected_df = circuits_df.select("circuitId", "circuitRef", "name", "location", "country", "lat", "lng", "alt")
```

```
display(circuits_selected_df)
```

Table

	circuitId	circuitRef	name	location	country	lat	lng
1	1	albert_park	Albert Park Grand Prix Circuit	Melbourne	Australia	-37.8497	144.96
2	2	sepang	Sepang International Circuit	Kuala Lumpur	Malaysia	2.76083	101.73
3	3	bahrain	Bahrain International Circuit	Sakhir	Bahrain	26.0325	50.510
4	4	catalunya	Circuit de Barcelona-Catalunya	Montmeló	Spain	41.57	2.2611
5	5	istanbul	Istanbul Park	Istanbul	Turkey	40.9517	29.40
6	6	monaco	Circuit de Monaco	Monte-Carlo	Monaco	43.7347	7.4205
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8	8	magny_cours	Circuit de Nevers Magny-Cours	Magny Cours	France	46.8642	3.1636
9	9	silverstone	Silverstone Circuit	Silverstone	UK	52.0786	-1.0169
10	10	hockenheimring	Hockenheimring	Hockenheim	Germany	49.3278	8.5658
11	11	hungaroring	Hungaroring	Budapest	Hungary	47.5789	19.248
12	12	valencia	Valencia Street Circuit	Valencia	Spain	39.4589	-0.33166
13	13	spa	Circuit de Spa-Francorchamps	Spa	Belgium	50.4372	5.9713
14	14	monza	Autodromo Nazionale di Monza	Monza	Italy	45.6156	9.2811

77 rows

```
circuits_selected_df = circuits_df.select(circuits_df.circuitId, circuits_df.circuitRef, circuits_df.name, circuits_df.location, ci
```

```
display(circuits_selected_df)
```

Table

	circuitId	circuitRef	name	location	country	lat	lng
1	1	albert_park	Albert Park Grand Prix Circuit	Melbourne	Australia	-37.8497	144.96
2	2	sepang	Sepang International Circuit	Kuala Lumpur	Malaysia	2.76083	101.73
3	3	bahrain	Bahrain International Circuit	Sakhir	Bahrain	26.0325	50.510
4	4	catalunya	Circuit de Barcelona-Catalunya	Montmeló	Spain	41.57	2.2611
5	5	istanbul	Istanbul Park	Istanbul	Turkey	40.9517	29.40
6	6	monaco	Circuit de Monaco	Monte-Carlo	Monaco	43.7347	7.4205
7	7	villeneuve	Circuit Gilles Villeneuve	Montreal	Canada	45.5	-73.522
8	8	magny_cours	Circuit de Nevers Magny-Cours	Magny Cours	France	46.8642	3.1636
9	9	silverstone	Silverstone Circuit	Silverstone	UK	52.0786	-1.0169
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11	11	hungaroring	Hungaroring	Budapest	Hungary	47.5789	19.248
12	12	valencia	Valencia Street Circuit	Valencia	Spain	39.4589	-0.33166
13	13	spa	Circuit de Spa-Francorchamps	Spa	Belgium	50.4372	5.9713
14	14	monza	Autodromo Nazionale di Monza	Monza	Italy	45.6156	9.2811

77 rows

```
circuits_selected_df = circuits_df.select(circuits_df["circuitId"], circuits_df["circuitRef"], circuits_df["name"],
                                          circuits_df["location"], circuits_df["country"], circuits_df["lat"], circuits_df["lng"], circuits_d
```

```
from pyspark.sql.functions import col
```

```
#circuits_selected_df = circuits_df.select(col("circuitId"), col("circuitRef"), col("name"), col("location"), col("country").alias(
```

```
display(circuits_selected_df)
```

Table

	circuitId	circuitRef	name	location	country	lat	lng
1	1	albert_park	Albert Park Grand Prix Circuit	Melbourne	Australia	-37.8497	144.96
2	2	sepang	Sepang International Circuit	Kuala Lumpur	Malaysia	2.76083	101.73
3	3	bahrain	Bahrain International Circuit	Sakhir	Bahrain	26.0325	50.510
4	4	catalunya	Circuit de Barcelona-Catalunya	Montmeló	Spain	41.57	2.2611
5	5	istanbul	Istanbul Park	Istanbul	Turkey	40.9517	29.40
6	6	monaco	Circuit de Monaco	Monte-Carlo	Monaco	43.7347	7.4205
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8	8	magny_cours	Circuit de Nevers Magny-Cours	Magny Cours	France	46.8642	3.1636
9	9	silverstone	Silverstone Circuit	Silverstone	UK	52.0786	-1.0169
10	10	hockenheimring	Hockenheimring	Hockenheim	Germany	49.3278	8.5658
11	11	hungaroring	Hungaroring	Budapest	Hungary	47.5789	19.248
12	12	valencia	Valencia Street Circuit	Valencia	Spain	39.4589	-0.33166
13	13	spa	Circuit de Spa-Francorchamps	Spa	Belgium	50.4372	5.9713
14	14	monza	Autodromo Nazionale di Monza	Monza	Italy	45.6156	9.2811

77 rows

Step3 - Rename the columns as required

```
from pyspark.sql.functions import lit
```

```
circuits_renamed_df = circuits_selected_df \
  .withColumnRenamed("circuitId", "circuit_id") \
  .withColumnRenamed("circuitRef", "circuit_ref") \
  .withColumnRenamed("lat", "latitude") \
  .withColumnRenamed("lng", "longitude") \
  .withColumnRenamed("alt", "altitude") \
  .withColumn("data_source", lit(v_data_sourec)) \
  .withColumn("file_date", lit(v_file_date))
```

```
display(circuits_renamed_df)
```

Table

	circuit_id	circuit_ref	name	location	country	latitude	lon
1	1	albert_park	Albert Park Grand Prix Circuit	Melbourne	Australia	-37.8497	
2	2	sepang	Sepang International Circuit	Kuala Lumpur	Malaysia	2.76083	
3	3	bahrain	Bahrain International Circuit	Sakhir	Bahrain	26.0325	
4	4	catalunya	Circuit de Barcelona-Catalunya	Montmeló	Spain	41.57	
5	5	istanbul	Istanbul Park	Istanbul	Turkey	40.9517	
6	6	monaco	Circuit de Monaco	Monte-Carlo	Monaco	43.7347	
7	7	villeneuve	Circuit Gilles Villeneuve	Montreal	Canada	45.5	
8	8	magny_cours	Circuit de Nevers Magny-Cours	Magny Cours	France	46.8642	
9	9	silverstone	Silverstone Circuit	Silverstone	UK	52.0786	
10	10	hockenheimring	Hockenheimring	Hockenheim	Germany	49.3278	
11	11	hungaroring	Hungaroring	Budapest	Hungary	47.5789	
12	12	valencia	Valencia Street Circuit	Valencia	Spain	39.4589	
13	13	spa	Circuit de Spa-Francorchamps	Spa	Belgium	50.4372	
14	14	monza	Autodromo Nazionale di Monza	Monza	Italy	45.6156	

77 rows

Step4 - Add ingestion date to dataframe

```
#from pyspark.sql.functions import current_timestamp
```

```
#circuits_final_df = circuits_renamed_df.withColumn("ingestion_date", current_timestamp())
```

```
circuits_final_df = add_ingestion_date(circuits_renamed_df)
```

```
display(circuits_final_df)
```

Table

	circuit_id	circuit_ref	name	location	country	latitude	lon
1	1	albert_park	Albert Park Grand Prix Circuit	Melbourne	Australia	-37.8497	
2	2	sepang	Sepang International Circuit	Kuala Lumpur	Malaysia	2.76083	
3	3	bahrain	Bahrain International Circuit	Sakhir	Bahrain	26.0325	
4	4	catalunya	Circuit de Barcelona-Catalunya	Montmeló	Spain	41.57	
5	5	istanbul	Istanbul Park	Istanbul	Turkey	40.9517	
6	6	monaco	Circuit de Monaco	Monte-Carlo	Monaco	43.7347	
7	7	villeneuve	Circuit Gilles Villeneuve	Montreal	Canada	45.5	
8	8	magny_cours	Circuit de Nevers Magny-Cours	Magny Cours	France	46.8642	
9	9	silverstone	Silverstone Circuit	Silverstone	UK	52.0786	
10	10	hockenheimring	Hockenheimring	Hockenheim	Germany	49.3278	
11	11	hungaroring	Hungaroring	Budapest	Hungary	47.5789	
12	12	valencia	Valencia Street Circuit	Valencia	Spain	39.4589	
13	13	spa	Circuit de Spa-Francorchamps	Spa	Belgium	50.4372	
14	14	monza	Autodromo Nazionale di Monza	Monza	Italy	45.6156	

77 rows

suppose we need to add another column environment

-- not part of project, just additional info

```
#from pyspark.sql.functions import lit
```

```
#circuits_final_df = circuits_renamed_df.withColumn("ingestion_date", current_timestamp()) \
#    .withColumn("environment", lit("production"))
```

```
#display(circuits_final_df)
```

Step5 - Write data to datalake as parquet

```
circuits_final_df.write.mode("overwrite").format("delta").saveAsTable("f1_processed.circuits")
```

```
%sql
SELECT * FROM f1_processed.circuits
```

Table

	circuit_Id	circuit_ref	name	location	country	latitude	lon
1	1	albert_park	Albert Park Grand Prix Circuit	Melbourne	Australia	-37.8497	
2	2	sepang	Sepang International Circuit	Kuala Lumpur	Malaysia	2.76083	
3	3	bahrain	Bahrain International Circuit	Sakhir	Bahrain	26.0325	
4	4	catalunya	Circuit de Barcelona-Catalunya	Montmeló	Spain	41.57	
5	5	istanbul	Istanbul Park	Istanbul	Turkey	40.9517	
6	6	monaco	Circuit de Monaco	Monte-Carlo	Monaco	43.7347	
7	7	villeneuve	Circuit Gilles Villeneuve	Montreal	Canada	45.5	
8	8	magny_cours	Circuit de Nevers Magny-Cours	Magny Cours	France	46.8642	
9	9	silverstone	Silverstone Circuit	Silverstone	UK	52.0786	
10	10	hockenheimring	Hockenheimring	Hockenheim	Germany	49.3278	
11	11	hungaroring	Hungaroring	Budapest	Hungary	47.5789	
12	12	valencia	Valencia Street Circuit	Valencia	Spain	39.4589	
13	13	spa	Circuit de Spa-Francorchamps	Spa	Belgium	50.4372	
14	14	monza	Autodromo Nazionale di Monza	Monza	Italy	45.6156	

77 rows

```
display(dbutils.fs.mounts())
```

Table

	mountPoint	source	encryptionType
1	/databricks-datasets	databricks-datasets	
2	/mnt/formula1dl1204/presentati...	abfss://presentation@formula1dl1204.dfs.core.windows.n...	

3	/mnt/formula1dl204/demo	abfss://demo@formula1dl1204.dfs.core.windows.net/	
4	/mnt/formula1dl1204/processed	abfss://processed@formula1dl1204.dfs.core.windows.net/	
5	/Volumes	UnityCatalogVolumes	
6	/mnt/formula1dl1204/raw	abfss://raw@formula1dl1204.dfs.core.windows.net/	
7	/databricks/mlflow-tracking	databricks/mlflow-tracking	
8	/databricks-results	databricks-results	
9	/databricks/mlflow-registry	databricks/mlflow-registry	
10	/Volume	DbfsReserved	
11	/mnt/formula1dl1204/demo	abfss://demo@formula1dl1204.dfs.core.windows.net/	
12	/volumes	DbfsReserved	
13	/	DatabricksRoot	
14	/volume	DbfsReserved	

14 rows

```
%sql
DESC  DATABASE EXTENDED f1_processed;
```

Table

	database_description_item	database_description_value
1	Catalog Name	spark_catalog
2	Namespace Name	f1_processed
3	Comment	
4	Location	dbfs:/mnt/formula1dl1204/processed
5	Owner	root
6	Properties	

6 rows

```
#display(spark.read.parquet(f"{processed_folder_path}/circuits"))
```

```
#df = spark.read.parquet(f"{processed_folder_path}/circuits")
```

```
%sql
select * from f1_processed.circuits;
```

Table

	circuit_Id	circuit_ref	name	location	country	1.2 latitude	1.2 lon
1	1	albert_park	Albert Park Grand Prix Circuit	Melbourne	Australia	-37.8497	
2	2	sepang	Sepang International Circuit	Kuala Lumpur	Malaysia	2.76083	
3	3	bahrain	Bahrain International Circuit	Sakhir	Bahrain	26.0325	
4	4	catalunya	Circuit de Barcelona-Catalunya	Montmeló	Spain	41.57	
5	5	istanbul	Istanbul Park	Istanbul	Turkey	40.9517	
6	6	monaco	Circuit de Monaco	Monte-Carlo	Monaco	43.7347	
7	7	villeneuve	Circuit Gilles Villeneuve	Montreal	Canada	45.5	
8	8	magny_cours	Circuit de Nevers Magny-Cours	Magny Cours	France	46.8642	
9	9	silverstone	Silverstone Circuit	Silverstone	UK	52.0786	
10	10	hockenheimring	Hockenheimring	Hockenheim	Germany	49.3278	
11	11	hungaroring	Hungaroring	Budapest	Hungary	47.5789	